Database Systems – ISYS2014

Vietnam Tourism Database – User Guide

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System requirements:

Component	Requirement
Operating system	Curtin Linux
Database management system	MySQL 8.4.0
Programming language	Python 3+
Python dependency	mysql-connector-python
Additional files	SQL scripts, CSV datasets, Part5.py

Accessing MySQL:

Open the terminal on the Curtin Linux system and navigate to the directory containing the assignment files:

```
cd /path/to/assignment_dir/
```

Connect to the MySQL database using the following username:

```
mysql -u dsuser -p
```

Enter the following password when prompted:

```
userCreateSQL
```

Creating the database and tables:

Within the MySQL command-line client, execute:

```
SOURCE Part2_1.sql
```

This script will create the database 'Phan_23028920', its tables, their primary keys, foreign keys, and other constraints. The implementation can be verified with:

```
SHOW TABLES;
```

Ensure the four CSV files are present in the same directory, then execute the following command to load the tables with data:

```
SOURCE Part2_2.sql
```

Execute the following statements to verify that the data has been loaded:

```
SELECT COUNT(*) FROM ArrivalByNationality;
SELECT COUNT(*) FROM ExpenditureByCategory;
```

Execute the following statement to implement the advanced database features:

```
SOURCE Part4_1.sql
```

This will create the following features:

1. insNewArrivalByNationality (Procedure)
2. after_ArrivalByNationality_update (Trigger)
3. YoY_ArrivalByNationality (View)

Using the database:

The eight predefined SQL queries are stored in 'Part3.sql' and can be executed through:

```
SOURCE Part3.sql
```

Individual queries may also be copied and pasted into the MySQL console.

The file 'Part4.sql' contains statements which will demonstrate the stored procedure, trigger, and view.

```
SOURCE Part4_2.sql
```

Keep in mind that these statements may modify the database records.

The following operations are demonstrated:

1. Successful insertion and failed insertion with transaction rollback
2. Updating an arrival count and inspecting its audit record
3. Querying year-on-year changes for recorded nationalities

Using the Python application:

Exit the MySQL console and return to the terminal. Ensure that you are still in the assignment directory. Create and activate a Python virtual environment:

```
python3 -m venv .venv
source .venv/bin/activate
```

Install the required Python dependencies:

```
pip install -r requirements.txt
```

Start the Python program:

```
python3 Part5.py
```

Enter the same username and password as above to access the MySQL server.

Below are the six different operations the user can take within the program.

Option	Operation	User input
1	View provincial tourism turnover	Year
2	View tourist arrivals by nationality	Nationality
3	Compare arrivals by transport between two years	Two years
4	View regional tourism turnover	Year
5	Insert a nationality arrival record	Year, nationality, count
0	Exit the application	None

Select an option by entering the number and pressing Enter. Different operations may require different input data. The program will construct the queries, execute them, and display the results to the user.

The user can exit by entering '0'. This will exit the main menu and close the MySQL connection.